

Just Talk, and Sticky Notes will be Created: Towards Collaborative Dialogue in Face-to-Face Workshops the Experiment with a Generative AI-Based

Gippeum Lee*

Interaction Design

Seoul Women's University

Seoul, Republic of Korea

delightleechristian@gmail.com

Namchoon Park

Industrial Design

Seoul Women's University

Seoul, Republic of Korea

ncpark@swu.ac.kr

Abstract

Taking visible notes is essential for enhancing productivity and creativity in workshops. Facilitators or participants often engage in dialogues while simultaneously jotting down opinions or capturing emerging ideas. However, focusing on note-taking can cause them to miss comments or lose track of the discussion. This raises the question: “Does note-taking for better collaboration hinder shared understanding and disrupt dialogues?” To explore this, we developed a tool that automates note-taking by displaying sticky notes in real-time on a digital whiteboard. We investigated: “Can generative AI tools preserve the value of note-taking while maintaining focus on dialogues in collaborative workshops?” The experiment showed that the tool increased participant engagement in dialogues, confirming its utility. However, it also highlighted potential benefits and drawbacks of automating note-taking. We discuss ways to address limitations and prevent losses caused by the automation of note-taking with generative AI.

CCS Concepts

• Human-centered computing; • Human computer interaction (HCI); • Empirical studies in HCI;

Keywords

Generative AI, Automated Note-Taking Tool, Collaborative Workshops, Empirical Study

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*Corresponding Author

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1 INTRODUCTION AND BACKGROUND

In collaborative workshops where participants take turns sharing their stories and gaining collective insights, it is crucial to document and visualize the dialogue in a way that everyone can see. Such documentation not only preserves content that might otherwise fade away, allowing participants to revisit and remember it, but also externalizes their contributions, offering opportunities to review their own opinions and those of others [1–4]. One of the most common methods of documenting dialogue is by using sticky notes [5, 6]. Facilitators often write down participants' ideas on sticky notes during the dialogue and place them on a visible surface, such as a whiteboard or a flat wall, or ask participants to write down their own thoughts directly [7].

According to Cognitive Load Theory (CLT), cognitive load arises in collaborative settings due to various factors. These include information shared among participants, the use of multiple communication channels, the construction and generation of new information, explaining or arguing positions, executing various procedures, task interruptions, and distractions [8, 9]. While a certain level of cognitive load is inevitable, it is crucial to regulate unnecessary cognitive load in the collaborative process to ensure effective engagement and performance [8].

Although note-taking by facilitators or participants is nearly indispensable for running a workshop, it requires significant cognitive effort, particularly in the complex, interaction-rich environment of a workshop. Facilitators, for instance, must engage participants through questions, manage time, and simultaneously document meaningful insights and opinions. This multitasking imposes a heavy cognitive load [10]. Participants, too, often find themselves note-taking their voiced opinions or writing down emerging ideas while simultaneously engaging in dialogue with others. However, focusing on the act of note-taking may lead them to miss out on others' contributions or lose track of the discussion flow [11–13]. In other words, the act of note-taking can divide participants' visual and cognitive attention, making it difficult for them to stay focused on the dialogue. The act of note-taking can divert participants' visual attention and cognitive focus, potentially hindering their ability to fully engage in the conversation [8].

This observation raises an important question: “Does note-taking for better collaboration inadvertently hinder shared understanding among team members and disrupt the continuity of dialogue?”

To address this concern, we sought to explore the potential of automating note-taking in a way that relieves the burden on facilitators and participants, while preserving its role as a tool

for collective memory and a stimulus for creative thinking. One study suggested that, during design thinking workshops, the note-taking process—such as using sticky notes—should be digitized and automated to allow participants to focus more on the dialogue. This functionality was proposed for integration into a digital agent [14].

Recently, advancements in generative AI have made it possible to process and summarize human language effectively. To address the challenges associated with manual note-taking, we developed a tool that automates note-taking using generative AI. In online environments, automated tools have already been developed to capture discussions based on spoken content during meetings or workshops. Chandrasegaran et al., who developed TalkTraces, a tool that captures and visualizes audio information in real-time using speech-to-text conversion and natural language processing technologies to extract topics from meeting content [15]; and Asthana et al., who designed and evaluated a meeting summarization system that leverages generative AI [16].

However, face-to-face environments present unique characteristics, such as more spontaneous dialogues and the use of physical tools and spaces, that differ from online settings [17, 18]. Therefore, it is essential to explore the effectiveness and limitations of automated note-taking tools in the context of face-to-face workshops. Ultimately, this study is guided by the following central question: *“In collaborative workshops conducted in face-to-face environments, can generative AI tools automate note-taking in a way that allows participants to focus on the dialogue while effectively preserving the importance of note-taking?”* Through the process of addressing this question, this study aims to make the following contributions: (1) introduce a prototype that enables participants in face-to-face workshops to immerse themselves in dialogue without interruptions caused by note-taking, through the automation of documentation, (2) conduct qualitative analyses of the tool’s effects, losses, and limitations by experimenting with the prototype in actual workshops, drawing on the experiences of both participants and facilitators, and (3) propose and discuss design considerations for automated note-taking tools, focusing on aspects such as time for reflection, the ability to edit sensitive information, error recovery, ease of use, and decision-making facilitation.

2 METHOD

This study involves the development of a prototype for an automated note-taking tool and its application in a real workshop setting to evaluate participants’ experiences.

2.1 Prototype

The prototype was developed using the no-code tool Bubble IO, utilizing its plugins for converting voice data to text and connecting APIs. Participants’ voices were transcribed into text at 60-second intervals, and the transcribed text was summarized using OpenAI’s GPT-4. The summarized content was then shared in real-time on a digital whiteboard in the form of sticky notes. Facilitators were provided with features to edit, delete, or add new sticky notes to supplement or modify the AI-generated content.

2.2 Workshop

2.2.1 Session Design. The workshop was designed to compare traditional note-taking methods and automated note-taking tools. It consists of a total of three sessions. In Session 1, participants and facilitators were asked to note down their opinions or ideas together without using any automated note-taking tools. In Sessions 2 and 3, automated note-taking tools were used; in Session 2, brief notes were provided, while in Session 3, detailed notes were provided. The provided notes could be viewed in real time. After sufficient sticky notes were generated in all three sessions, participants gathered to reflect on the conversation and consolidate the opinions and ideas that emerged. In the experiment, it was necessary to ensure a high utilization of sticky notes throughout the entire process of the actual workshop. Therefore, the Wizard of Oz technique was applied. Two facilitators worked on pasting the sticky notes generated by the developed tool onto sticky notes of the same color in Figma. The facilitators and participants were regularly provided with sticky notes every 1 minute and 30 seconds, in accordance with the timing of when the tool generated the notes. This experiment was conducted during a total of three workshops in September 2024, with each workshop lasting approximately 3 hours and 30 minutes.

2.2.2 Workshop Topics and Participants. To test an automated note-taking tool, We selected the topic ‘Memorizing App Service for Generation Z’, with three specific themes for each session: calendar app, study note app, and memory management app service. We chose these themes because they can be grouped under similar topics while possessing distinct features, allowing for the testing of three different aspects. The topics for each session were arranged randomly to control for any impact a specific theme might have on the experimental results.

Through a pre-survey, We recruited 12 Generation Z participants who have experience using calendar apps, study note apps, and memory management services. These participants were randomly selected. Additionally, three facilitators with facilitation experience were also randomly selected through a pre-survey.

Each workshop included one facilitator and four participants, totaling 15 individuals. All participants were offered a reward of 70 dollars. Since none of the workshop participants had previous experience using automated note-taking tools, we conducted a pre-training session to adequately explain how to use the automated tools and the differences from traditional methods. We also provided guidance on the differences in outcomes based on the specificity of the summaries.

2.2.3 Evaluation and Analysis. After each session, both the facilitator and participants completed evaluations on usability, usefulness, and reliability, measured on a 1 to 5-point Likert scale. After the workshop concluded, a comparative evaluation of the traditional method and the automated tool method was conducted, along with in-depth interviews.

The collected survey data and interview results were used to analyze the impact of the automated note-taking tool on the workshop experience from multiple perspectives. In particular, the analysis focused on whether the automated tool increased participants’ engagement in dialogue and facilitated idea generation, as well as how

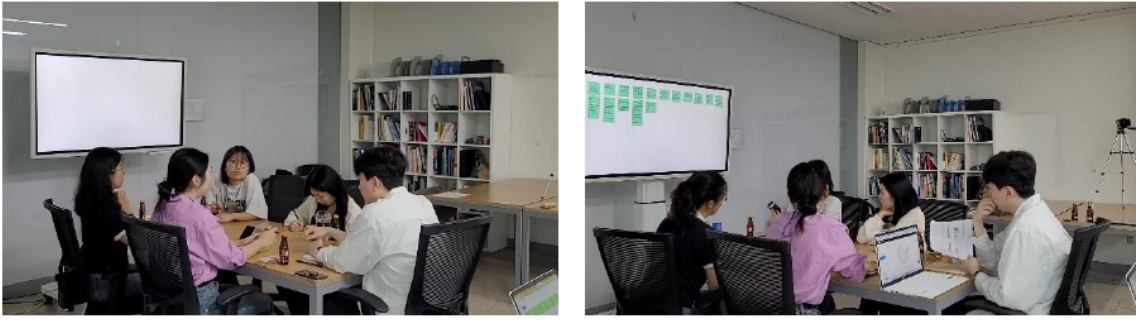


Figure 1: Photos from the workshop experiment (Left: Session 1, workshop without the use of an automated note-taking tool; Right: Session 2 or 3, workshop using the automated note-taking tool).

the differences in summarization methods affected the tool’s usefulness and usability. Additionally, the smoothness of the facilitator’s operation was also evaluated.

3 FINDINGS

The results were organized into two phases: the phase where dialogue and note-taking occur and the phase where sticky notes are utilized. From the perspective of the tool’s utility, specific effects and losses were identified, while from the perspective of usability, participants’ experiences regarding the timing of sticky note provision and errors in content were documented.

3.1 Phase of Dialogue and Note-Taking

3.1.1 The tool that automates note-taking aids dialogue and progress but limits participants’ thinking. As anticipated, we found that automating the note-taking phase helped participants immerse themselves in dialogue during the sessions. One participant shared, “Both coming up with my own ideas and taking notes while listening to others’ ideas require focus. In the first session, trying to do both made me feel tense and drained. But with automated note-taking, I only had to focus on the dialogue” (P7). Additionally, participants reported that in the sessions using the automated tool, they were able to make more eye contact, concentrate on others’ ideas, and provide more feedback while actively engaging with one another. “Since all we had to do was talk, we could make eye contact and really listen to each other’s voices. That made the discussions more active” (P8). Four participants, in particular, mentioned that the automated tool was especially helpful when they found it challenging to note down their own thoughts. “I usually find it hard to write down my ideas in an organized way, so note-taking was a burden for me. It was much easier when I could just focus on the conversation” (P3).

On the other hand, in the sessions where participants manually noted down their ideas, facilitators often waited until the dialogue ended before asking participants to write down the ideas discussed. During this process, some participants found it cumbersome to recall and document what they had previously said. “Having to revisit and write down what I’d already talked about felt tedious and inconvenient” (P1).

These responses highlight that, while previous studies have described sticky notes as “a low-tech and non-specialized design material that anyone can intuitively use to make creative contributions” [19], they can present challenges when used as a tool for note-taking in the context of memorizing dialogue. However, our findings confirm that tools that automate note-taking can significantly alleviate the burden of documentation for participants, enabling them to immerse themselves in dialogue in a more comfortable and free-flowing atmosphere.

The facilitators confirmed that they were able to use their time efficiently during the workshop session where note-taking was automated. Facilitators F2 and F3 exceeded the given time during the session where participants were required to take notes directly. However, in the session with automated note-taking, they did not exceed the time limit. “After the conversation ended, asking them to write took quite a while as well. I think that really helped reduce a lot of time” (F2). Particularly, F1, who had less facilitation experience, found it confusing to manage note-taking simultaneously during traditional design workshops. They reported that the automated tool significantly helped them. “I didn’t have to worry about whether the facilitator should also organize the sticky notes; it really helped that this tool handled it all for me” (F1).

Although the automation of note-taking provided assistance to both participants and facilitators, there were also opinions that this led to a loss of thinking time. “I think I was able to organize my thoughts and generate ideas while writing, but since AI does it all, I feel like I lost that opportunity” (P4).

In workshops where participants took notes directly, they could organize their thoughts through the act of writing and generate new ideas based on that. This relates to the benefits gained from the process of producing sticky notes, where ambiguous and implicit meanings in participants’ minds can be creatively interpreted and reinterpreted to propose new opinions or ideas [8]. However, automated note-taking tools skip this writing process, which may cause participants to lose the opportunity to organize, interpret, and reinterpret their thoughts.

While the automated note-taking tool takes the form of sticky notes, it can eliminate the possibility of using the sticky notes themselves as a means to facilitate participants’ thinking during the writing process.

3.1.2 In the Process of Automating Note-Taking, Timing of Summaries Must Be Appropriate, and Content Errors Can Cause Concern. In the session where note-taking was automated, participants primarily focused on the conversation, but they occasionally checked to see if sticky notes were being generated. Specifically, they would verify immediately after speaking to see if sticky notes had been created, and once they observed that the generative AI was summarizing in real-time, they built trust and then refocused on the dialogue. Participants commented on this as follows: *“It seemed like it was done pretty well when I checked in between”* (P1). *“I felt that the responsiveness was highest when it was exactly matched to the end of the conversation”* (P8).

However, when participants saw that a sticky note had been generated before their speech ended, they felt a lack of completeness in the notes and experienced confusion. In response to this, participants said, *“When it suddenly appeared while we were talking, it felt like my words weren’t fully heard, and it just organized and posted it”* (P7). *“When sticky notes were generated while someone was speaking, I thought it shouldn’t be cut off here, and when it seemed like nothing was posted at all, I wondered if the AI had missed something”* (P8). In such cases, they lost trust in the automated note-taking tool and made efforts to remember what they had said. *“I thought it might not capture all my words completely. So I felt I should write down keywords whenever I remembered them”* (P2).

Unlike humans, participants held a mental model that machines provide notes based on a structured algorithm. They understood that while a person can be inspired and take notes during a conversation, a machine does not get inspired in the same way. Therefore, the timing for the automated note-taking tool to provide notes should be appropriate, ideally when the participant’s speech has ended. Otherwise, they may perceive it as an ‘error’ in processing by the algorithm, leading to a loss of trust in the tool.

3.2 Phase of Utilizing Sticky Notes

3.2.1 It Allows for the Capture of Ideas Without Value Judgments. The tool that automates note-taking summarized and provided all content from the conversation, allowing the tool to note down every utterance from participants, including those deemed unimportant, which opens the possibility for those ideas to lead to new insights. One participant responded that all ideas potentially hold value and that it is essential to keep all possibilities open. *“I think this could be important information or it might not be. But honestly, if this becomes a sticky note later, we don’t know what idea it might turn into”* (P1).

Additionally, while participants often went through a value judgment process to decide whether their opinions were valuable and only wrote down what they deemed meaningful, the automated note-taking tool recorded everything, resulting in a wider range of ideas being generated. *“In fact, when we were writing on the sticky notes, there were many instances where I judged that something was too detailed and didn’t need to be written down. But when we use AI, it proceeds through an algorithm regardless of our perceived importance, so it feels like we can manage more data and it was noted down without missing anything”* (P4).

Participants tend to refine their thoughts and ideas, but they also believe that ‘unrefined, raw, and odd’ ideas can develop into good

ideas, which should be adequately shared. Therefore, it was noted that a ‘neutral’ automated tool could fulfill their desire for more creative exchanges.

3.2.2 Understanding the content of sticky notes generated by the automated tool requires additional effort. In sessions utilizing sticky notes that participants noted down themselves, they organized their notes by attaching them and categorizing ideas by priority or emphasizing key insights. Since participants clearly remembered their own sticky notes, they could quickly confirm connections with other opinions and group the notes accordingly.

However, in sessions using sticky notes generated by an automated tool, participants did not write the notes themselves, so they had to read through all the sticky notes to identify which content reflected their own contributions. One participant noted, *“There was the hassle of having to read everything again”* (P12).

Additionally, while the automated tool provided content in a consistent style, which was expected to make understanding the sticky notes easier, it actually posed challenges in comprehension. The inability to identify who authored the content based on handwriting or style made it difficult to trace the origin of the opinions. *“When reviewing the content, there were things I didn’t remember well. Since it wasn’t written in my own style, I found myself needing time to think about what the content was”* (P1).

In this regard, it was recognized that although the sticky notes generated by the automated tool were based on participants’ spoken contributions, a process of individually recognizing the sticky notes was still necessary since they were not written by the participants themselves. Furthermore, it became clear that a means of distinguishing speakers was needed for participants to better identify their contributions.

3.2.3 Types of Errors in Automated Note-Taking Tools and Reactions. Four types of errors emerged in the automated note-taking tool: omission of entire opinions, omission of parts of opinions, misinterpretation of entire opinions, and misinterpretation of parts of opinions. Participants expressed confusion when recognizing these errors while reviewing the sticky notes, and they attempted to correct the content by inferring missing or inaccurately noted information. They noted that detailed sticky notes were easier to recall, while those that were summarized briefly made it relatively difficult to infer meaning when errors occurred. *“It was summarized so briefly that even one or two typos could make it really difficult to understand the context”* (P3). *“I had high expectations when AI was summarizing. I was curious about how it would organize what we discussed, but it turned out to be just one line, like ‘Happy about typing fast.’ So I ended up feeling a bit disappointed. I had explained so much, but it was processed so casually as something trivial, which felt a bit empty. I realized that what I had explained earlier was really important, and it left me feeling regretful”* (P7).

From the participants’ responses, it was confirmed that they held high expectations for the automated note-taking tool, and when errors occurred, they felt disconcerted. It highlighted the need for a mechanism to recover from these errors to be provided alongside the tool.

3.2.4 Ethical Issues of Note-Taking Private Information. During the workshop, participants mentioned personal experiences, such as

incidents involving their boyfriends, which the automated note-taking tool recorded. One participant expressed feeling uncomfortable about this. Another participant responded, *“If it were a workshop with sensitive topics that included many personal and sensitive comments, I think there would need to be more precautions regarding the use of AI”* (P4).

Thus, it was confirmed that when dealing with sensitive subjects, there have been instances where personal stories were publicly noted, causing participants to feel embarrassed. This indicates a need for improvements in handling such situations.

4 DISCUSSIONS

We were able to identify four key considerations when providing automated note-taking in workshops: (i) allowing participants to have time to think for themselves, (ii) enabling the editing of personal private information, (iii) providing a means to recover from errors, and (iv) ensuring that alternatives and methods facilitate utilization and decision-making.

4.1 Allowing Participants to Have Time to Think for Themselves

Automated tools need to expand beyond simply generating real-time notes and should provide participants with time to think for themselves. In traditional note-taking methods, the act of noting down served as a moment to pause the conversation and organize thoughts; however, it was noted that this process was lacking in automated note-taking. Sometimes silence is necessary to hear someone else’s opinion. Silence allows for quiet idea generation and creates a persuasive visual framework that everyone can intuitively follow [12]. One participant responded, *“If we just talk without time to think, we often end up with disorganized thoughts. It would be better if we could take a moment to review what has been noted and organize our thoughts in between”* (P4).

To address this, automated tools should be combined with methodologies that intentionally insert moments for participants to review or organize the noted content at specific intervals.

This can be interpreted from the perspective of cognitive load as proposed by Sweller [20]. Sweller categorized cognitive load into three types: intrinsic cognitive load, extraneous cognitive load, and germane cognitive load. In this study, efforts were made to reduce the extraneous cognitive load in the note-taking process of facilitators and participants. However, since writing on sticky notes involves thinking and structuring ideas to understand and solve problems, it also constitutes germane cognitive load, which is essential for workshops. Through the process of germane cognitive load, participants can understand new concepts, establish connections, and actively engage in learning while participating in the collaborative process of the workshop. Therefore, automated tools should not completely replace manual note-taking but rather be integrated with methodologies that intentionally incorporate moments for participants to review or organize recorded content themselves at specific points during the workshop. In this process, it could be designed to pause note generation, allowing participants to create new notes themselves.

4.2 Allowing Editing of Personal Information

During the workshop, participants mentioned personal experiences, such as incidents involving their boyfriends, which the automated note-taking tool recorded. One participant expressed feeling uncomfortable about this. Thus, it was confirmed that when dealing with sensitive subjects, there have been instances where personal stories were publicly noted, causing participants to feel embarrassed. This indicates a need for improvements in handling such situations.

4.3 Providing Recovery Options for Errors

Machines can make errors at any time. While the reduction of the burden of note-taking through automated tools is a positive aspect, the summaries provided by these tools have the potential for inaccuracies, and there should still be methods in place to recover from such errors.

In this regard, many participants mentioned that even with real-time note-taking of conversations, indicating who spoke could help the speaker correct any inaccuracies. Additionally, to prepare for instances where summaries are missing, there could be provisions for reviewing the full transcripts of the conversation to provide any omitted content.

4.4 Facilitate Utilization and Decision-Making

The goal of the design workshop is to generate creative ideas through dialogue and make decisions based on those ideas. To achieve this, note-taking should not merely serve as stored information but should be presented in a way that can be easily utilized in the decision-making process. Therefore, automated tools should summarize the key points of the conversation and organize the context and related information for each idea, allowing participants to quickly review and draw conclusions.

Since participants did not engage in note-taking themselves, they faced difficulties in categorizing and recombining the sticky notes. To address this, automated tools should offer features that categorize conversation content in real-time by topic or structure it based on importance. If categorization is done automatically, both facilitators and participants can significantly reduce the time spent reviewing or utilizing the notes.

5 CONCLUSION & FUTURE WORKS

The insights presented in this paper illustrate the effects, losses, and limitations observed when automating the note-taking of participants’ dialogues in face-to-face workshops. Automated note-taking enhances the quality of dialogue in design workshops, opening up new possibilities for both participants and facilitators. However, for such automation to be effectively utilized, it is essential to design the process in a way that allows participants to retain control over the generated sticky notes. This study proposes several methods to achieve this, emphasizing the need to consider time for reflection, the ability to edit personal information, error recovery options, ease of use, and decision-making flexibility in the design of automated note-taking tools.

And we particularly emphasize the importance of allowing sufficient time for reflective thinking, which should be regarded as a meaningful and essential cognitive load rather than merely a burden.

Therefore, we stress the need to develop approaches that stimulate participants' thinking and facilitate deeper discussions, rather than fully automating the recording of collaborative conversations.

This study was conducted with Generation Z participants, who are familiar with digital tools and AI concepts. While this demographic is well-suited for evaluating modern technology, real-world workshops often involve participants from diverse age groups and backgrounds. Therefore, future research should include users with varying levels of technological proficiency to gain broader insights into the accessibility and usability of AI-driven note-taking tools.

Additionally, this study focused more on the role of note-taking rather than exploring the diverse applications of sticky notes, as discussed in previous research. However, future studies should also consider how sticky notes are utilized in different contexts and their interactions with the note-taking process. Notably, this study separated the note-taking phase from the utilization phase, potentially overlooking the benefits of categorizing sticky notes in real-time during the note-taking process. Future research could explore parallel scenarios where note-taking and utilization occur simultaneously, incorporating diverse workshop processes for more in-depth studies.

Although the scope of this research is limited, we hope this study will stimulate discussions on designing and utilizing automated tools in various scenarios where active dialogue and note-taking must occur concurrently. In particular, we encourage further discussions on how to effectively design and implement automation in collaborative environments where real-time interactions and documentation play a crucial role.

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